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Unlikelihood of Condorcet's paradox in a large society

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Abstract. We provide intuitive, formal, and computational evidence that in a large society Condorcet's paradox (the intransitivity of social preference obtained by pairwise vote) can hardly occur. For that purpose, we compare two models of social choice, one based on voting and another one based on summing individual cardinal utilities, expressed either in reals, or integers. We show that in a probabilistic model with a large number of independent individuals both models, almost surely, provide the same decision results. This implies that Condorcet's and Borda's methods tend to give the same decisions as the number of voters increases. Therefore, in the model with a large number of voters, the transitivity of the Borda preference is inherent in a majority preference as well.

1 Introduction

1.1 Social choice with respect to cardinal and ordinal utility

As early as in the last quarter of the 18th century, Borda, Condorcet, and Laplace mathematically investigated election by vote (see Black 1958; Young 1988; McLean and Urken 1994). Borda described cases when voting implies intuitively unacceptable results. Condorcet discovered the intransitivity of majoritarian preference, the fact known as Condorcet's paradox. Both authors developed new methods to overcome this problem with voting, which posed further problems. Laplace provided some mathematical arguments to back up Borda's model. These studies, having questioned the faithfulness of election systems, have had numerous successors and thus have given birth to a new scientific branch, the social choice theory.

Borda and Condorcet founded two different standpoints, which continue to be the two principal approaches to the problem. Borda's count is based on summing ranks of candidates in individual schedules, which are cardinal utilities. Laplace considered a model with real utilities as a logical development of Borda's model. Condorcet's count uses pairwise votes, which are ordinal data. Recall that cardinal and ordinal approaches to utility differ, respectively, in taking, or not taking, into account the degree of preference expressed numerically. Ordinal models of collective choice deal with the count of voters who prefer one alternative to an other, whereas in cardinal models the total cardinal utility of an alternative is determined from its individual cardinal utilities. For instance, the cardinal approach is used in welfare economics, where a social welfare is understood to be a sum of individual utilities.

Under the cardinal approach, alternatives are evaluated by numbers, that is, projected on the real line, so that the collective preference is always transitive. However, the methodological foundations of cardinal models are rather questionable. A direct inquiry for cardinal utility is possible if only the judgment standards are well established (grades at exams, scores at athletic competitions); otherwise cardinal utilities from different individuals are hardly commensurable. Another problem is that cardinal utility permits a voter to overrate favorite alternatives and to underrate their competitors.

The ordinal approach, including a majority rule, is usually considered to be 'more objective' and better substantiated than the cardinal one. On the other hand, the intransitivity of preference discovered by Condorcet is generally inherent in all ordinal models. Thus having a rigorous methodological basis, the ordinal approach lacks the universal solvability inherent in the cardinal approach. In a general formulation, the unsolvability of the ordinal approach to social choice has been proved by Arrow (1951).

However, contrary to Condorcet's predictions, the intransitivity of social preference is seldom observed in real voting. Moreover, all democratic systems are based on different forms of voting, ignoring the unsolvability of the ordinal approach. So we have an apparent inconsistency between theory and practice.

In this paper, we analyze the cause of this inconsistency. It turns out that Condorcet's predictions are true for a quite particular combinatorial case, when an 'average voter' votes for or against certain candidates with equal probability. The related analytical studies have been done by DeMeyer and Plott (1970), May (1971), Merril and Tideman (1991), Niemi and Weisberg (1968), Garman and Kamien (1968), Gehrlein and Fishburn (1976), and Gehrlein (1983). The results of computer simulation have been reported by Campbell and Tullock (1965), Klahr (1966), Jones et al. (1995), and Van Deemen (1996).

The assumption of equal probability of voting for or against candidates means that these candidates have equal merits, a circumstance which is possible theoretically, but can hardly ever occur. Much more frequently, candidates have different merits, reflected by the vote asymmetry. We show that in such a case the probability of intransitive social preference vanishes as the number of voters increases. From this we conclude that it is very unlikely to obtain Condorcet's paradox in a real vote.

We have to emphasize that our results cannot be applied literally to coalition voting, since the independence assumption does not hold.

1.2 Condorcet's paradox

Condorcet (1785) noted that the choice of the best candidate should take into account more information than simple votes for the best candidate. He suggested using the information about whole preference orders (voting schedules, in D. Black's (1958) terminology) of electors. According to Condorcet, the individual preferences should be aggregated into the group preference, and then the best alternative from the group preference should be selected as the group choice.

Condorcet suggested that the majoritarian preference obtained from voting on every pair of alternatives should be used as the group preference. However, he discovered that a majority preference relation can be intransitive (cyclic), the fact known as Condorcet's paradox. To illustrate this situation, consider candidates A, B, and C and three electors with the following preferences:

$$\text{Preferences of electors} \quad \uparrow \begin{array}{ccc} A & B & C \\ B & C & A \\ C & A & B \end{array} \quad \Rightarrow \quad A \overset{2:1}{\succ} B \overset{2:1}{\succ} C \overset{2:1}{\succ} A.$$

Then the candidate A is preferred to candidate B by 2 votes against 1, the candidate B is preferred to candidate C by 2 votes against 1, and the candidate C is preferred to candidate A by 2 votes against 1. The resulting cyclic collective preference $A \succ B \succ C \succ A$ makes the choice of the best candidate impossible.

Condorcet estimated the probability of an intransitive collective preference. For this purpose he assumed that majority vote gives all possible antisymmetric orderings on n candidates (where for every two candidates A or B, either $A \succ B$, or $B \succ A$) with equal probabilities (Black 1958, p. 173). Such ordering are associated with antisymmetric matrices like

$$\begin{array}{c|cccc} & A & B & C & \cdots \\ \hline A & 0 & 1 & -1 & \cdots \\ B & -1 & 0 & 1 & \cdots \\ C & 1 & -1 & 0 & \cdots \\ \multicolumn{5}{l}{\cdots\cdots\cdots\cdots\cdots} \end{array} \quad \Rightarrow \quad A \succ B \succ C \succ A.$$

The number of degrees of freedom in such matrices is the number of elements above the main diagonal: $(n^2 - n)/2 = n(n-1)/2$, implying the total number of such matrices to be $2^{n(n-1)/2}$. Since the number of transitive (linear) orderings on n elements is $n!$, the fraction of intransitive preferences $[2^{n(n-1)/2} - n!]/[2^{n(n-1)/2}]$ increases rapidly as the number of candidates increases (Table 1).

Table 1 An increase in the fraction of intransitive preferences

Number of candidates	Number of possible majority preferences	
	Transitive	Intransitive
2	2	0
3	6	2
4	24	40
5	120	904

1.3 Borda's model

Unlike Condorcet's model, Borda's method of marks (which according to Black (1958) dates back to 1770) uses cardinal utilities, that is numerical evaluations of alternatives.

Borda also discovered that simple votes for the best candidate can result in an intuitively wrong choice. In particular, the winner can be the candidate which is least preferred by a majority. For example, consider 7 electors and three candidates A, B and C ordered as shown:

$$\text{Preferences of electors} \uparrow \begin{array}{ccccccc} A & A & A & B & B & C & C \\ B & B & B & C & C & B & B \\ C & C & C & A & A & A & A. \end{array}$$

Borda has suggested the following method of marks. Each elector $i = 1, \ldots, n$ evaluates candidate with p_{ij} points (marks). The best candidate gets m points, the next best $m - 1$ points, and so on. Better candidate then has a greater sum of marks

$$P_j = \sum_{i=1}^{n} p_{ij}, \quad j = 1, \ldots, m.$$

In our example, 1st place = 3 points, 2nd place = 2 points, 3rd place = 1 point. Then

$$P_A = 3 \times 3 + 1 \times 4 = 13$$

$$P_B = 3 \times 2 + 2 \times 5 = 16 \quad \text{(the best candidate)}$$

$$P_C = 3 \times 2 + 2 \times 2 + 1 \times 3 = 11.$$

As one can see, Borda's method of marks surmounts some shortcomings of a simple voting for the best candidate. However, it poses further questions, in particular: Do integer-valued marks represent cardinal utilities of individuals sufficiently adequately?

1.4 Laplace's model

The last question was answered by Laplace in 1795 (Black 1958).

He suggested that if individuals could really provide adequate estimates of alternatives by reals from the segment [0; 1] then the sum of individual utilities would be a trustworthy basis for a social preference (this approach is usual in welfare economics).

Specifically, Laplace supposed that each elector i can evaluate the candidate j by a real utility estimate $u_{ij} \in (0; m)$ (the greater the better), proportional to the degree of preference for the candidate. Then a better candidate has a greater sum of individual utilities

$$U_j = \sum_{i=1}^{n} u_{ij}, \quad j = 1, \dots, m.$$

Laplace assumed that (a) the electors are statistically independent, and (b) the *ordered* utility estimates of every individual are equidistributed in the interval $(0, m)$ (such an assumption is traditional in probability and statistics when nothing else is known). The latter assumption implies in particular that the distribution of cardinal utilities is independent on the ordinal preferences.

Under these assumptions the expectation of estimates of every elector has the integer-valued ratio $1 : 2 : \cdots : m$, as in Borda's model. Consequently, if the number of electors is large then, by the law of large numbers, the choice by Borda's model (by the sum of integer-valued marks) approximates the choice by Laplace's model (by the sum of real estimates):

$$P_j \xrightarrow[n \to \infty]{} U_j, \quad j = 1, \dots, m.$$

This means that under the equidistribution assumption for a large society the models of Borda and Laplace are almost for sure equivalent. One can conclude that the sums of ordinal ranks contain almost as much information as the sums of cardinal utilities.

1.5 About the given work

Under Laplace's assumptions, Tanguiane (1991, Chapter 4) derived distribution functions of the utility estimates of alternatives and tested the statistical hypothesis that Borda's count gives the same preference as Laplace's count. This result permits us, with a given fiducial probability, to replace Laplace's count (accurate but non-manageable) by Borda's count (inaccurate but manageable).

In this work we attempt the next step. We prove with a probability model that group decisions based on a pairwise majority vote or on summing individual cardinal utilities (in the form of either reals, or integers, or ranks) tend to be the same as the number of individuals increases. The only exception is the case when the vote ratio for pairs of alternatives is 50% : 50%. Since such an equilibrium is rather unlikely, we conclude that in most practical situations pairwise vote count and utility count are almost equivalent.

Let us explain the main idea at an intuitive level, in terms of tournaments. Imagine a chess competition with three participants A, B, and C. Due to chance, a weaker player can gain a victory over a stronger one, and the preference order on the set of players can have cycles like $A \succ B \succ C \succ A$. However, in multiple rounds, a stronger player wins more often, getting a higher total score of victories over weaker players. After a certain number of rounds, the preference order on the set of participants based on paired games will correspond to their real strength rating. (Recall that each chess professional is assigned a rating calculated with respect to his results in several tournaments.)

A pairwise vote for candidates A, B, C in a small group of voters is like a single tournament round with a risk of misrepresentation of real merits of the participants. This can result in intransitive cycles in the preference order. A vote in a large society, due to the error-vanishing effect, is similar to a multiple-round tournament where the merits of participants are revealed more accurately. If social merits of candidates are adequately reflected by the vote in an election, then the (latent) merit scale linearizes the resulting preference, as the rating scale linearizes the order of players in a multiple-round tournament.

The idea of the underlying 'objective' preference being distorted in individual preferences goes back to Condorcet who has assumed probabilistic deviations from some 'correct' vote. From this standpoint, the intransitivity of the majoritarian preference is no longer a paradox but rather an error to be corrected – the argument exploited by Condorcet and some other scholars.

Our approach can be explained with a reference to Black's (1958) model with single-peaked preferences. He proved the transitivity of a majority preference, having assumed an alignment of candidates such that each elector has a single-peaked preference with respect to this alignment. In our model, such an alignment is the latent social merit scale. In a large model, the deviations from this scale in individual preferences can be considered as 'errors' which become negligible as the number of individuals increases.

Note that, unlike Black, we do not restrict the domain of preference profiles (combinations of individual preferences). We consider all profiles under some probabilistic assumptions (independence of voters, inequality of the mean preference probability from 0.5, that is no equilibrium of votes). Then we show that the probability of the profiles which result in an intransitive social preference vanishes as the number of voters increases. We conclude that to avoid Condorcet's paradox in a large society, there is no need to introduce any domain restriction, since the domain 'restricts itself' almost for sure in the probabilistic sense under rather weak assumptions. That is, the profiles resulting in an intransitive social preference constitute an event with vanishing probability. Therefore, the trend displayed in Table 1 does not reflect the real risk of meeting an undesired preference profile.

In intermediate reasoning we make use of individual cardinal utilities. We assume that they do *exist* and have some probability distribution, but do not require their expressibility. This assumption is not crucial, since the question of existence of probability measures is trivial for finite models, which is our case (the number of individuals is finite, and individual utilities can be restricted to some finite grid).

Since the limit theorems of the paper are valid for arbitrary distributions, the results hold even if these distributions are not known. Here we follow the same line of reasonong as (Tanguiane 1991, 1994) while revising Arrow's paradox: We prove a property for every measure and then conclude that knowing the measure is not necessary to be sure that this property holds.

In Sect. 2, "Results", we consider a model with two alternatives. We prove that the decision obtained by voting and that obtained by summing individual cardinal utilities become the same as the number of individuals increases. The two decisions contradict each other with a significant probability only if the probability of vote for one candidate against another is equal to 0.5. This case corresponds to precisely equal social merits of the candidates which is a rare coincidence.

Then, we extend this result from a pair of alternatives to several alternatives. This gives the main statement of the paper: The social preferences based on ordinal or cardinal counts tend to become the same as the number of individuals increases. As a corollary, in a model with a large number of voters, the transitivity of the Borda preference is inherent in the preference based on pairwise vote.

In Sect. 3, "Numerical modeling and interpretation", we provide numerical evidence of the coincidence of results from voting and from summing individual utilities. The speed of convergence of the two methods of social decision making depends on several parameters. We classify distributions of individual cardinal utilities with respect to these parameters and interpret these classes with regard to the way of social behavior. The computed tables of fiducial probabilities for five principal types are given in the Appendix at the end of the paper.

We show with an example how these tables could be used to estimate the size of the society where a majority preference is transitive almost for sure (say, with the significance level 1%). For instance, this estimate can be used in planning a Gallup Poll of public opinion to specify the minimal representative group to be interviewed.

In Sect. 4, "Conclusions", we comment on the applicability of the results of this paper and illustrate some political implications with an example of recent presidential elections in Russia. Finally, we recapitulate the main statements of the paper.

2 Results

2.1 Voting versus summation of cardinal utilities in case of two alternatives

As in Laplace's model, assume that individual preferences are measured cardinally by reals from the segment $[0; 1]$ and that every alternative is evaluated by the sum of individual utilities. In such a case, the alternative that is determined as best for the group is not necessarily the one with most votes. We shall compare the decision by the sum of cardinal utilities with that by vote in a probabilistic model.

First, consider the case of n individuals and two alternatives A and B. Let every individual i, $i = 1, \dots, n$, have a utility function $u_i(x)$ which takes values from the segment $[0; 1]$. We say that A is *socially preferred* to B if

$$\sum_{i=1}^{n} u_i(A) \geq \sum_{i=1}^{n} u_i(B). \tag{1}$$

It means that we use the social welfare function $W(x) = \sum_{i=1}^{n} u_i(x)$.

Decompose the cardinal utility of every individual i, $i = 1, \dots, n$, into the ordinal part (= vote) and the cardinal residual (= degree of preference) by defining the following variables

$$\eta_i = \begin{cases} 1 & \text{if } A \succeq_i B \quad \Leftrightarrow \quad u_i(A) \geq u_i(B) \\ 0 & \text{if } A \prec_i B \quad \Leftrightarrow \quad u_i(A) < u_i(B) \end{cases}$$

$$\xi_i = |u_i(A) - u_i(B)|.$$

The condition that A is preferred to B by a majority is equivalent to $\sum_{i=1}^{n} \eta_i \geq \frac{n}{2}$, or

$$\sum_{i=1}^{n} (2\eta_i - 1) > 0$$

The condition (1) is equivalent to $\sum_{i=1}^{n} \eta_i \xi_i \geq \sum_{i=1}^{n} (1 - \eta_i) \xi_i$, or

$$\sum_{i=1}^{n} (2\eta_i - 1) \xi_i > 0.$$

Following Laplace, assume that η_i and ξ_i, $i = 1, \dots, n$ are independent random variables. This assumption means that (a) the individuals are independent, and (b) the degree of preference is not statistically influenced by the adherence to A or to B. (That is, we do not consider the cases when, for instance, *A*BBA fans are on the average more enthusiastic than *B*eatles fans).

In the sequel, the Bernoulli random variables η_i (votes) will be characterized by the probabilities

$$p_i = \mathsf{P}\{\eta_i = 1\}, \quad q_i = \mathsf{P}\{\eta_i = 0\} = 1 - p_i, \quad i = 1, \dots, n,$$

$$p = \frac{1}{n} \sum_{i=1}^{n} p_i, \quad q = \frac{1}{n} \sum_{i=1}^{n} q_i.$$

The continuous random variables ξ_i (degree of preference) will be characterized with the mathematical expectation and the variance

$$\mu_i = \mathsf{E}\xi_i, \quad \sigma_i^2 = \mathsf{V}\xi_i, \quad i = 1, \dots, n,$$

$$\mu = \frac{1}{n} \sum_{i=1}^{n} \mu_i, \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^{n} \sigma_i^2.$$

We shall also assume that the model is essentially probabilistic, that is, there is a coalition with a positive weight $\theta > 0$ with individuals whose preferences, both ordinal and cardinal, have a non-zero variances separable from 0 by $\varepsilon > 0$. In a general form this condition has been formulated by Tanguiane (1991, Chapter 7; see also 1994, p. 27).

Lemma 1 (Equality of large society decisions by vote and by summing individual cardinal utilities for two alternatives)**.** *Consider a model with two alternatives and a large number n of independent individuals. Assume that the ordinal preference η_i of every individual i is independent of the degree of preference ξ_i. Let a significant number of individuals have a probabilistic behavior, that is, let there exist $\theta > 0$ and $\varepsilon > 0$ such that $\varepsilon < p_i < 1 - \varepsilon$ and $\sigma_i^2 > \varepsilon$ for $i = 1, 2, \ldots < \theta n$. Then*

$$\mathsf{P}\{\text{Decision by vote} \neq \text{Decision by summing cardinal utilities}\}$$

$$= \mathsf{P}\left\{\sum_{i=1}^{n}(2\eta_i - 1) \leq 0 \text{ and } \sum_{i=1}^{n}(2\eta_i - 1)\xi_i > 0\right\} \tag{2}$$

$$+ \mathsf{P}\left\{\sum_{i=1}^{n}(2\eta_i - 1) > 0 \text{ and } \sum_{i=1}^{n}(2\eta_i - 1)\xi_i \leq 0\right\} \tag{3}$$

$$\approx \int_{\mathscr{A} \cup \mathscr{B}} e^{-(x^2+y^2)/2}\, dx\, dy \tag{4}$$

$$\xrightarrow[n\to\infty]{} \begin{cases} 0 & \text{if } p \neq 1/2 \\ \dfrac{\arctan \dfrac{\sigma}{\mu}}{\pi} & \text{if } p = 1/2, \end{cases}.$$

where

$$\mathscr{A} = \left\{(x, y) : x < \frac{1/2 - p}{\sqrt{pq}}\sqrt{n},\, y < \frac{2\mu}{\sigma}[\sqrt{pq}x + (p - 1/2)\sqrt{n}]\right\}$$

$$\mathscr{B} = \left\{(x, y) : x > \frac{1/2 - p}{\sqrt{pq}}\sqrt{n},\, y > \frac{2\mu}{\sigma}[\sqrt{pq}x + (p - 1/2)\sqrt{n}]\right\}.$$

The demonstration of Lemma 1 is found in Sect. 5.

Thus, for a large society, voting and summing cardinal utilities almost for sure give the same results. The only exception is the case $p = 1/2$, when these two methods are not equivalent, and the degree of inequivalence depends on the deviation coefficient of the individual cardinal utilities σ/μ. This result is quite natural: If the voting probability $p = 1/2$ then in a large society the votes are close to 50% : 50%, and adding weight coefficients to votes (cardinal utilities) may result in a different social preference. If the weight coefficients are all equal ($\sigma = 0$) then no change of decision can occur (both $\alpha, \beta \xrightarrow[n\to\infty]{} 0$).

If the weight coefficients deviate considerably from a small mean value, that is, $\sigma/\mu \gg 0$ then the measure of inconsistency of ordinal and cardinal methods can approach $1/2$ $\left(\text{since } \arctan \frac{\sigma}{\mu} \xrightarrow[\frac{\sigma}{\mu}\to\infty]{} \frac{\pi}{2}\right)$.

2.2 General case of several alternatives

In case of $m > 2$ alternatives, we have $m(m-1)/2$ unordered pairs of alternatives. By Lemma 1 for every positive $\delta > 0$ and every two alternatives A, B one can indicate the size of the society $n(A, B)$ such that the cardinal and ordinal decisions on A, B coincide with probability $1 - \delta$. Since the number of pairs of alternatives is finite, the probability that the preference as a whole will coincide is not less than $(1-\delta)^{m(m-1)/2}$ for the society with the number of individuals $n \geq \max_{A,B} n(A, B)$. We obtain the following theorem.

Theorem 1 (Equality of decisions of a large society by voting and by summing individual cardinal utilities for whole preferences). *Consider several alternatives and let the mean vote for every pair of them be* $p(A, B) \neq 1/2$, *that is, let all the alternatives have unequal merits. Then*

$$\mathsf{P}\{\text{Preference from pairwise voting} = \text{Preference from summing utilities}\} \xrightarrow[n\to\infty]{} 1.$$

Clearly, the trend established above is stronger for a smaller number of alternatives.

Since a cardinal preference is always transitive (as projected onto a real line), we obtain the following corollary.

Theorem 2 (Solution to the Condorcet paradox). *Consider several alternatives and let the mean vote for every pair of them be* $p(A, B) \neq 1/2$, *that is, all the alternatives have unequal merits. Then the fraction of cases when the social preference obtained from pairwise voting is not transitive vanishes as the number of individuals increases.*

With minor changes, our model can be applied to Borda's method of marks. Fix two alternatives and consider cardinal utilities ξ_i, with which individuals $i = 1 \ldots, n$ evaluate the two alternatives. In case of the method of marks, ξ_i takes equidistant values as $\frac{1}{m+1}, \ldots, \frac{m}{m+1}$. If the society is sufficiently large then the estimates ξ_i can be considered as random variables with a certain mean expectation and standard deviation. From Theorem 1 we obtain that, with a high probability, voting and Borda count provide the same social preference.

Theorem 3 (Equivalence of Borda and Condorcet count in a large society). *Consider several alternatives and let the mean vote for every pair of them* $p(A, B) \neq 1/2$ *(i.e., all the alternatives have unequal merits). Then*

$$\mathsf{P}\{\text{Preference from pairwise voting} = \text{Preference from Borda count}\} \xrightarrow[n\to\infty]{} 1.$$

In particular, in a large society Borda and Condorcet count tend to be equivalent.

3 Numerical modeling and interpretation

3.1 Convergence of decisions versus types of social behavior

Let us numerically trace the convergence of decisions by voting and by summing cardinal utilities for different types of individual utility distributions. Without loss of generality, we restrict our attention to the case of two alternatives (since Lemma 1 is generalizable to the case of an arbitrary finite set of alternatives).

According to Lemma 1 the result of voting contradicts the result of summing individual utilities with a probability $\alpha + \beta$ given in (8) and (9). One can see that this probability is determined by the following three factors:

- number of individuals n,
- asymmetry of vote $|p - 1/2|$, and
- the ratio σ/μ which characterizes the distribution of individual utilities.

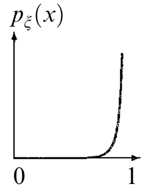
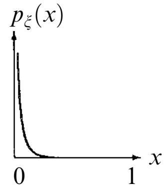
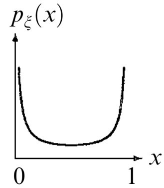
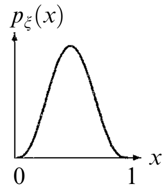
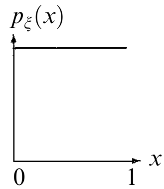
To comment on the last parameter, consider the distribution of the individual utility estimate ξ on the interval $(0; 1)$. Table 2 displays possible types of its density curve by means of beta-densities defined as follows

$$p_\xi(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 \le x \le 1,\ \alpha > 0,\ \beta > 0.$$

With only two parameters, α and β, it enables us to model various types of social behavior which we explain conventionally referring to psychological character types:

- $0 < \alpha < 1,\ \beta > 1$. In this situation most voters have definitive preferences expressed by the highest estimates.
  An individual with such a distribution of utilities would be passionate for or against something with no intermediate states. Such extremities are usually inherent in a choleric temperament, as noted in the second column of the table.
- $\alpha > 1,\ 0 < \beta < 1$. In this situation most voters have weak preferences, with no strong desires. At the individual level, the indifference is inherent in a phlegmatic temperament.
- $0 < \alpha < 1,\ 0 < \beta < 1$. This type of social response is said to be relaxing, with a fraction of passionate voters of the first type and a fraction of relaxed voters of the second type.
  In case of a single individual, abrupt switching to indifference is inherent in a melancholic, or even hysteric temperament.
- $\alpha > 1,\ \beta > 1$. In this case utilities are located around an 'objective' estimate of an alternative. This type of preference expression is said to be moderate. The realistic estimates are inherent in equilibrated individuals with a sanguine temperament.

Table 2 Distributions of individual utilities

Social response	Personification	α	β	Density curve	μ	σ	σ/μ
Extreme	Choleric	9.90	0.10		0.99	0.03	0.0303
Indifferent	Phlegmatic	0.10	9.90		0.01	0.03	3.0000
Relaxing	Melancholic	0.10	0.10		0.50	0.46	0.9129
Moderate	Sanguine	4.00	4.00		0.50	0.17	0.3333
Mixed	Unpredictable	1.00	1.00		0.50	0.29	0.5774

- $\alpha = \beta = 1$. In this case the beta distribution degenerates into the equal distribution in the interval $(0; 1)$. It corresponds to a mixed society where all the voter types are present.
  Being inherent in the behavior of a single individual, such a variety of preference expression makes the voting of a person unpredictable.

Obviously, the density curves can be modified by adjusting the parameters α and β within the limits listed. For instance, for the curves of the first class $(0 < \alpha < 1,\ \beta > 1)$, the closer α to 0 and the greater β, the closer is the density curve to the right-hand vertical $\xi = 1$. This means that the social response is getting more extreme.

The delimitation of certain classes is rather conventional. In fact, the density peak in the 'Moderate' class can be close to 0 or to 1, being interpreted as 'indifferent' or 'extreme' social response. A similar comment can be made on the density minimum in the 'Relaxing' class.

As predicted by Lemma 1, the probability that the result of voting contradicts the decision based on summing utilities vanishes as the number of voters increases. The more asymmetric the vote (the greater $|p - 1/2|$) and the less scattered the preference intensities (the smaller σ/μ), the more salient is this trend. This follows from (8) and (9).

The related computational results are displayed in five tables computed with the formulas (8) and (9) for σ/μ as in Table 2.

3.2 *Application to Gallup Polls*

Now let us illustrate how to use the tables with an example.

Example 1 (Estimation of the minimal group with a transitive majoritarian preference). Suppose that we want to rank 5 candidates for president from the results of a Gallup Poll. By virtue of Theorem 2, if the number of polled persons is sufficiently large then the majoritarian preference is transitive with a high probability. The question is: How many persons should be polled in order to obtain a transitive majoritarian preference with a probability better than 0.99?

As already mentioned, if the society had a cardinal preference, it would be transitive. Therefore, we shall estimate the minimal group with the same cardinal and ordinal preference.

Assume *a priori* that the candidates are not equal in merits, so that we estimate the asymmetry in every pairwise 'voting' to be greater than 2%. Assume also that there are no great tensions in the society, so that we expect a moderate social response with the ratio $\sigma/\mu = 1/3$ as in Table 2.

Thus we restrict our attention to the third column ($|\bar{p} - \frac{1}{2}| = 0.02$) of Table 4. We find that for a single pairwise vote (for any two candidates) with 3000 voters and more, voting and summing cardinal utilities give the same result with a probability better than 0.99.

Since we have $m = 5$ candidates, the number of their pairs $\dfrac{m(m-1)}{2} = 10$. Therefore, we have to guarantee that the results from all the 10 pairwise votes are equivalent to that from summing individual utilities. The event that the whole preference is the same is obviously the intersection of the 10 events for pairwise comparisons.

Recall that for the probability of a joint event we have $\mathsf{P}(AB) \geq \mathsf{P}(A)\mathsf{P}(B)$. Therefore, to make the probability of the required event for the whole preference greater than or equal to 0.99 we can use the probability for a single pair of alternatives $0.99^{1/10} \approx 0.999$ which is attained for 6000 individuals.

Thus, under the assumptions made, in order to obtain a transitive majoritarian preference on 5 candidates, at least 6000 individuals should be interviewed.

Since the tables have several entries, they can be used to test hypotheses about parameters of the model. For instance, a cycle in the majoritarian preference of a large society implies one of the following causes: (a) the individuals are essentially dependent; (b) the mean vote probability p is close to 0.5; (c) the distribution of preference intensities ξ corresponds to the 'indifferent' type. These implications can be formulated as a statistical test with a given significance level.

4 Conclusions

From the notation of the proof of Lemma 1, since σ is limited, the ratio σ/μ is large only if μ is small. This occurs only if the utility estimates are close to 0, meaning weak preference, almost total indifference. This corresponds to the 'Indifferent' type of social response in Table 2 (or to the 'Moderate' response biased to the left with the same interpretation). Thus in most practical cases (not simultaneously vote symmetry and indifferent social response) decisions based on voting and on summing individual utilities provide close results. For a large society, reservations should be made only for the case of symmetric vote in of indifferent society.

It should be emphasized that Condorcet's paradox remains relevant for small groups. Since our model assumes the probabilistic independence of individuals, a large society means a large number of *independent* individuals. A few large coalitions with members sharing the same preference do not meet this condition. Such a model is a model with a few participants with the weight coefficients, corresponding to the size of coalitions.

Independence may be regarded as the most consistent realization of the idea of probability. Therefore, our results relate with reservations to a non-independent probabilistic model, as demonstrating some kind of a general trend.

Besides, our results illustrate the opinion that cardinal measurements are not very important in processing ordinal data. In fact, we show that voting leads to the same results as summing cardinal utilities, implying that most essential information is already inherent in ordinal preferences.

Finally, it follows that the inaccuracy of certain election procedures should not play a decisive role in case of a large society. In particular, a majority vote will reveal the same preference as more sophisticated and 'precise' methods of election that are based on cardinal utilities.

For instance, consider the presidential election in Russia for Jelzin against Sjuganow on June 16, 1996. In spite of a small vote asymmetry (34% and 32% of votes, respectively) we can expect that the social response was not indifferent (since the political programs of the candidates differed significantly). From our standpoint we can be quite sure that even a hypothetical voting system which takes into account the degree of individual preferences would give the same election result.

In conclusion we summarize the main statements of the paper.

1. Our model is not based on any domain restriction on the individual preference profiles implying the transitivity of a majority preference. We consider all profiles but show under rather weak assumptions that the probability of an intransitive collective preference is vanishing as the number of individuals increases (Condorcet has considered equiprobable collective preferences which, as one can see, is not relevant to real social choice).
2. For this purpose we assume the following
   - Independence of individuals (following Condorcet, Laplace),
   - Independence of ordinal and cardinal parts in individual preferences (following Borda, Laplace),
   - Asymmetry of pairwise votes (Tangian).
3. In a large society, a different choice with respect to voting or summing individual utilities is statistically not significant. The solvability of a large ordinal model of social choice (transitivity of a majoritarian preference) follows from the *existence* of the dual cardinal model, not requiring its explicit knowledge.
4. In a small group, a different choice with respect to voting and summing individual utilities is statistically significant. Testing the difference requires additional information about the distribution of preference probabilities.
5. A different choice with respect to voting and summing individual utilities is caused by the following:
   - a high deviation coefficient σ/μ (since $\sigma < 1$, the expectation μ must be small, that is, individuals are indifferent);
   - almost symmetrical votes (p is close to $1/2$).
6. One should not worry about the imperfection of election systems based on voting:
   - The risk that an intransitive preference emerges is negligible.
   - Election based on voting is almost as accurate as complex systems with cardinal utilities.

5 Proof of Lemma 1

We shall prove that $\sum_{i=1}^{n}(2\eta_i - 1)\xi_i$ is asymptotically normally distributed by applying the central limit theorem to the sums $\sum_{i=1}^{n} \eta_i\xi_i$ and $\sum_{i=1}^{n} \xi_i$. For this purpose it suffices to verify that the Lyapunov ratios

$$L(\xi) = \frac{\sum_{i=1}^{n} \mathsf{E}|\xi_i - \mathsf{E}\xi_i|^3}{(\sum_{i=1}^{n} \mathsf{V}\xi_i)^{3/2}},$$

$$L(\xi\eta) = \frac{\sum_{i=1}^{n} \mathsf{E}|\eta_i\xi_i - \mathsf{E}(\eta_i\xi_i)|^3}{[\sum_{i=1}^{n} \mathsf{V}(\eta_i\xi_i)]^{3/2}}$$

are vanishing as $n \to \infty$ (Loève 1977).

At first we prove three auxiliary inequalities. Consider a Bernoulli random variable $\eta \sim B(p), \varepsilon < p < 1 - \varepsilon$, and an independent random variable $0 \leq \xi \leq 1$ with $\mathsf{E}\xi = \mu$ and $\mathsf{V}\xi = \sigma^2 > \varepsilon$, where $\varepsilon > 0$. Since $0 \leq \xi \leq 1$, and $0 \leq \eta \leq 1$, we have $0 \leq \mathsf{E}(\xi) \leq 1$ and $0 \leq \mathsf{E}(\eta\xi) \leq 1$. Consequently, $|\xi - \mathsf{E}(\xi)| \leq 1$ and $|\eta\xi - \mathsf{E}(\eta\xi)| \leq 1$. Hence,

$$\mathsf{E}|\xi - \mathsf{E}(\xi)|^3 \leq \mathsf{E}[\xi - \mathsf{E}(\xi)]^2 = \mathsf{V}(\xi), \tag{5}$$

$$\mathsf{E}|\eta\xi - \mathsf{E}(\eta\xi)|^3 \leq \mathsf{E}[\eta\xi - \mathsf{E}(\eta\xi)]^2 = \mathsf{V}(\eta\xi). \tag{6}$$

By virtue of independence of the variables we have

$$\begin{aligned}
\mathsf{V}(\eta\xi) &= \mathsf{E}[\eta\xi - \mathsf{E}(\eta\xi)]^2 \\
&= p\mathsf{E}(1 \cdot \xi - p\mu)^2 + q(0 - p\mu)^2 \\
&= p\mathsf{E}\xi^2 - 2p^2\mu^2 + p^3\mu^2 + qp^2\mu^2 \\
&= p\mathsf{E}\xi^2 - 2p^2\mu^2 + p^2\mu^2(p+q) \\
&= p\mathsf{E}\xi^2 - p^2\mu^2 \\
&= p[\mathsf{E}\xi^2 - (p+q)\mu^2 + q\mu^2] \\
&= p\sigma^2 + pq\mu^2 \\
&\geq \varepsilon^4.
\end{aligned} \tag{7}$$

Now we estimate the Lyapunov ratios.

$$\begin{aligned}
L(\xi) &= \frac{\sum_{i=1}^{n} \mathsf{E}|\xi_i - \mathsf{E}(\xi_i)|^3}{[\sum_{i=1}^{n} \mathsf{V}(\xi_i)]^{3/2}} \quad \text{substitute (5)} \\
&\leq \frac{\sum_{i=1}^{n} \mathsf{V}(\eta_i)}{[\sum_{i=1}^{n} \mathsf{V}(\eta_i)]^{3/2}} \\
&= \left(\sum_{i=1}^{n} \sigma_i^2\right)^{-1/2} \\
&\leq (\theta n\varepsilon)^{-1/2} \\
&\xrightarrow[n\to\infty]{} 0,
\end{aligned}$$

$$L(\xi\eta) = \frac{\sum_{i=1}^{n} \mathsf{E}|\eta_i\xi_i - \mathsf{E}(\eta_i\xi_i)|^3}{[\sum_{i=1}^{n} \mathsf{V}(\eta_i\xi_i)]^{3/2}} \quad \text{substitute (6)}$$

$$\leq \frac{\sum_{i=1}^{n} \mathsf{V}(\eta_i\xi_i)}{[\sum_{i=1}^{n} \mathsf{V}(\eta_i\xi_i)]^{3/2}}$$

$$= \left(\sum_{i=1}^{n} \mathsf{V}(\eta_i\xi_i)\right)^{-1/2} \quad \text{substitute (7)}$$

$$\leq (\theta n \varepsilon^4)^{-1/2}$$

$$\xrightarrow[n\to\infty]{} 0.$$

Thus $\sum_{i=1}^{n}(2\eta_i - 1)\xi_i$ is asymptotically normally distributed. Note that the limit normal density depends only on the mean expectation and the mean variance of all the random variables. Therefore, the limit normal density is the same as if all the random variables had the mean expectation and the mean variance. Thus, we estimate the probability α assuming that $p_i = p$, $\mu_i = \mu$, and $\sigma_i = \sigma$ for all $i = 1, \ldots, n$.

By α denote the probability (2) of the event when B is preferred to A by a majority while A being socially preferred. Note that each relevant combination of $\eta_1, \ldots, \eta_n$ contains k ones, $k < n/2$, corresponding to k voters who prefer A to B, and $n - k$ zeros, corresponding to $n - k$ voters who do not prefer A to B. At the same time, the total degree of preference of the k voters is superior to that of all other voters. Taking into account the independence of all the variables, we obtain

$$\alpha = \sum_{k=0}^{n/2} C_n^k p^k q^{n-k} \mathsf{P}\left\{\sum_{i=0}^{k} \xi_i - \sum_{i=k+1}^{n} \xi_i > 0\right\}.$$

By the central limit theorem, the random variable $\sum_{i=0}^{k} \xi_i - \sum_{i=k+1}^{n} \xi_i$ has an asymptotically normal distribution with the expectation $[k - (n-k)]\mu = (2k - n)\mu$ and the standard deviation $\sigma\sqrt{n}$. Therefore, for large n we have

$$\alpha \approx \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{n/2} C_n^k p^k q^{n-k} \int_{(0-(2k-n)\mu)/\sigma\sqrt{n}}^{+\infty} e^{-(y^2/2)}\, dy$$

$$= \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{n/2} C_n^k p^k q^{n-k} \int_{-\infty}^{((2k-n)\mu)/\sigma\sqrt{n}} e^{-(y^2/2)}\, dy.$$

Applying the central limit theorem to the sum in the above formula, and taking into account that after the transition to continuous axis, $x = (k - np)/\sqrt{npq}$ whence $k = np + x\sqrt{npq}$, we obtain

$$\alpha \approx \frac{1}{2\pi}\int_{-\infty}^{(n/2-np)/\sqrt{npq}} dx\, e^{-x^2/2} \int_{-\infty}^{((2np+2x\sqrt{npq}-n)\mu)/\sigma\sqrt{n}} e^{-(y^2/2)}\, dy$$

$$= \frac{1}{2\pi}\int_{-\infty}^{(1/2-p)\sqrt{n/pq}} dx \int_{-\infty}^{(2\mu/\sigma)[x\sqrt{pq}+(p-1/2)\sqrt{n}]} dy\, e^{-((x^2+y^2)/2)}. \tag{8}$$

Similarly, denote by β the probability (3) and

$$\beta = \mathsf{P}\left\{\sum_{i=1}^{n}(2\eta_i - 1) > 0 \text{ and } \sum_{i=1}^{n}(2\eta_i - 1)\xi_i \le 0\right\}$$

$$\approx \frac{1}{2\pi}\int_{(1/2-p)\sqrt{n/pq}}^{\infty} dx \int_{(2\mu/\sigma)[x\sqrt{pq}+(p-1/2)\sqrt{n}]}^{\infty} dy\, e^{-((x^2+y^2)/2)}. \tag{9}$$

One can see that α is the integral of the function $e^{-(x^2+y^2)/2}$ over the two-dimensional domain

$$\mathscr{A} = \left\{(x, y) : x < \frac{1/2 - p}{\sqrt{pq}}\sqrt{n},\, y < \frac{2\mu}{\sigma}[\sqrt{pq}x + (p - 1/2)\sqrt{n}]\right\},$$

and β is the integral of the same function over the domain

$$\mathscr{B} = \left\{(x, y) : x > \frac{1/2 - p}{\sqrt{pq}}\sqrt{n},\, y > \frac{2\mu}{\sigma}[\sqrt{pq}x + (p - 1/2)\sqrt{n}]\right\}$$

shown in Fig. 1. Let us investigate what happens to these two domains as $n \to \infty$.

If $p - 1/2 < 0$ (Fig. 1a) then the vertical line which restricts $\mathscr{A}$ and $\mathscr{B}$ moves to the right, and the inclined line moves downward. Since the probability density $e^{-(x^2+y^2)/2}$ is concentrated around point $(0, 0)$, we obtain $\alpha \xrightarrow[n\to\infty]{} 0$ and $\beta \xrightarrow[n\to\infty]{} 0$.

If $p - 1/2 > 0$ (Fig. 1b) then the vertical line which restricts $\mathscr{A}$ and $\mathscr{B}$ moves to the left, and the inclined line moves upward. Again, $\mathscr{A}$ and $\mathscr{B}$ move away from the point $(0, 0)$, whence $\alpha \xrightarrow[n\to\infty]{} 0$ and $\beta \xrightarrow[n\to\infty]{} 0$.

If $p - 1/2 = 0$ (Fig. 1c) then the vertical line which restricts $\mathscr{A}$ and $\mathscr{B}$ is fixed at $x = 0$, and the inclined line is always $y = \frac{\mu}{\sigma}x$. Since the probability density $e^{-(x^2+y^2)/2}$ is circularly symmetric, the sum of integrals (8) and (9) is equal to the ratio of the double area of $\mathscr{A}$ to that of the whole plane $\mathscr{R}^2$. From geometric reasons we obtain:

$$\alpha + \beta = 2\alpha = 2\frac{\pi/2 - \arctan\frac{\mu}{\sigma}}{2\pi} = \frac{\arctan\frac{\sigma}{\mu}}{\pi},$$

as required. *Q.E.D.*

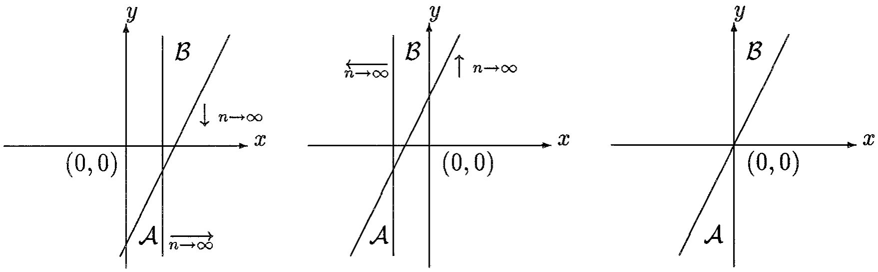


Fig. 1a–c Domains $\mathscr{A}$ and $\mathscr{B}$ in three cases. **a)** Case $p < 1/2$; **b)** Case $p > 1/2$; **c)** Case $p = 1/2$

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Appendix

Tables of probabilities

Table 3 Probabilities of the event that the decision by voting contradicts that obtained by summing individual utilities with $\sigma/\mu = 0.0303$

Number of voters n	Asymmetry of vote $\lvert\bar{p} - \frac{1}{2}\rvert$									
	0.0	0.01	0.02	0.03	0.04	0.05	0.1	0.2	0.3	0.4
10	0.0096	0.0096	0.0095	0.0094	0.0093	0.0092	0.0079	0.0040	0.0007	0.0
20	0.0096	0.0096	0.0094	0.0093	0.0090	0.0087	0.0064	0.0015	0.0	0.0
30	0.0096	0.0095	0.0094	0.0091	0.0087	0.0083	0.0052	0.0006	0.0	0.0
40	0.0096	0.0095	0.0093	0.0089	0.0085	0.0079	0.0042	0.0002	0.0	0.0
50	0.0096	0.0095	0.0092	0.0088	0.0082	0.0075	0.0034	0.0	0.0	0.0
60	0.0096	0.0095	0.0091	0.0086	0.0079	0.0071	0.0028	0.0	0.0	0.0
70	0.0096	0.0095	0.0091	0.0085	0.0077	0.0068	0.0022	0.0	0.0	0.0
80	0.0096	0.0094	0.0090	0.0083	0.0074	0.0064	0.0018	0.0	0.0	0.0
90	0.0096	0.0094	0.0089	0.0082	0.0072	0.0061	0.0015	0.0	0.0	0.0
100	0.0096	0.0094	0.0089	0.0080	0.0070	0.0058	0.0012	0.0	0.0	0.0
200	0.0096	0.0092	0.0082	0.0067	0.0050	0.0035	0.0001	0.0	0.0	0.0
300	0.0096	0.0090	0.0075	0.0056	0.0036	0.0021	0.0	0.0	0.0	0.0
400	0.0096	0.0088	0.0070	0.0046	0.0026	0.0012	0.0	0.0	0.0	0.0
500	0.0096	0.0087	0.0064	0.0039	0.0019	0.0007	0.0	0.0	0.0	0.0
600	0.0096	0.0085	0.0059	0.0032	0.0014	0.0004	0.0	0.0	0.0	0.0
700	0.0096	0.0083	0.0055	0.0027	0.0010	0.0002	0.0	0.0	0.0	0.0
800	0.0096	0.0082	0.0050	0.0022	0.0007	0.0001	0.0	0.0	0.0	0.0
900	0.0096	0.0080	0.0046	0.0019	0.0005	0.0001	0.0	0.0	0.0	0.0
1000	0.0096	0.0078	0.0043	0.0015	0.0003	0.0	0.0	0.0	0.0	0.0
2000	0.0096	0.0064	0.0019	0.0002	0.0	0.0	0.0	0.0	0.0	0.0
3000	0.0096	0.0052	0.0008	0.0	0.0	0.0	0.0	0.0	0.0	0.0

4000	0.0096	0.0043	0.0003	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5000	0.0096	0.0035	0.0001	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6000	0.0096	0.0029	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7000	0.0096	0.0023	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8000	0.0096	0.0019	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9000	0.0096	0.0015	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10000	0.0096	0.0013	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
...										
∞	0.0096	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 4 Probabilities of the event that the decision obtained by voting contradicts that obtained by summing individual utilities with $\sigma/\mu = 1/3$

Number of voters n	Asymmetry of vote $\lvert\bar{p} - \frac{1}{2}\rvert$									
	0.0	0.01	0.02	0.03	0.04	0.05	0.1	0.2	0.3	0.4
10	0.1024	0.1022	0.1017	0.1009	0.0996	0.0981	0.0859	0.0462	0.0100	0.0001
20	0.1024	0.102	0.101	0.0991	0.0967	0.0936	0.0708	0.0192	0.0008	0.0
30	0.1024	0.1018	0.1002	0.0975	0.0938	0.0893	0.0583	0.0080	0.0	0.0
40	0.1024	0.1016	0.0994	0.0958	0.0910	0.0852	0.0480	0.0033	0.0	0.0
50	0.1024	0.1014	0.0987	0.0942	0.0883	0.0813	0.0396	0.0014	0.0	0.0
60	0.1024	0.1014	0.0979	0.0927	0.0857	0.0775	0.0326	0.0005	0.0	0.0
70	0.1024	0.1011	0.0972	0.0911	0.0832	0.074	0.0269	0.0002	0.0	0.0
80	0.1024	0.1009	0.0965	0.0896	0.0807	0.0706	0.0221	0.0	0.0	0.0
90	0.1024	0.1007	0.0957	0.0881	0.0783	0.0673	0.0182	0.0	0.0	0.0
100	0.1024	0.1005	0.0951	0.0866	0.0760	0.0642	0.0150	0.0	0.0	0.0
200	0.1024	0.0986	0.0882	0.0732	0.0563	0.0401	0.0021	0.0	0.0	0.0
300	0.1024	0.0968	0.0819	0.0618	0.0417	0.0250	0.0003	0.0	0.0	0.0
400	0.1024	0.0950	0.076	0.0523	0.0309	0.0156	0.0	0.0	0.0	0.0
500	0.1024	0.0933	0.0705	0.0442	0.0229	0.0098	0.0	0.0	0.0	0.0
600	0.1024	0.0915	0.0654	0.0373	0.0169	0.0061	0.0	0.0	0.0	0.0
700	0.1024	0.0899	0.0607	0.0315	0.0125	0.0038	0.0	0.0	0.0	0.0
800	0.1024	0.0882	0.0563	0.0266	0.0093	0.0023	0.0	0.0	0.0	0.0
900	0.1024	0.0865	0.0523	0.0225	0.0069	0.0015	0.0	0.0	0.0	0.0
1000	0.1024	0.0849	0.0485	0.0190	0.0051	0.0009	0.0	0.0	0.0	0.0
2000	0.1024	0.0705	0.0230	0.0035	0.0002	0.0	0.0	0.0	0.0	0.0
3000	0.1024	0.0585	0.0109	0.0006	0.0	0.0	0.0	0.0	0.0	0.0
4000	0.1024	0.0485	0.0051	0.0001	0.0	0.0	0.0	0.0	0.0	0.0
5000	0.1024	0.0403	0.0024	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6000	0.1024	0.0334	0.0011	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7000	0.1024	0.0277	0.0005	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8000	0.1024	0.0230	0.0002	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9000	0.1024	0.0191	0.0001	0.0	0.0	0.0	0.0	0.0	0.0	0.0

10000	0.1024	0.0158	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
...										
∞	0.1024	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 5 Probabilities of the event that the decision obtained by voting contradicts that obtained by summing individual utilities with $\sigma/\mu = 0.5774$

Number of voters n	Asymmetry of vote $\lvert\bar{p} - \frac{1}{2}\rvert$									
	0.0	0.01	0.02	0.03	0.04	0.05	0.1	0.2	0.3	0.4
10	0.1667	0.1664	0.1656	0.1644	0.1627	0.1605	0.1428	0.0835	0.0243	0.0011
20	0.1667	0.1661	0.1646	0.162	0.1584	0.1539	0.1204	0.0392	0.0031	0.0
30	0.1667	0.1658	0.1635	0.1596	0.1543	0.1477	0.1015	0.0185	0.0004	0.0
40	0.1667	0.1656	0.1624	0.1572	0.1502	0.1416	0.0856	0.0087	0.0	0.0
50	0.1667	0.1653	0.1613	0.1549	0.1463	0.1359	0.0722	0.0041	0.0	0.0
60	0.1667	0.165	0.1602	0.1526	0.1424	0.1303	0.0609	0.0019	0.0	0.0
70	0.1667	0.1647	0.1592	0.1503	0.1387	0.125	0.0514	0.0009	0.0	0.0
80	0.1667	0.1645	0.1581	0.1481	0.1351	0.1199	0.0434	0.0004	0.0	0.0
90	0.1667	0.1642	0.1571	0.1459	0.1315	0.115	0.0366	0.0002	0.0	0.0
100	0.1667	0.1639	0.1561	0.1437	0.1281	0.1103	0.0309	0.0001	0.0	0.0
200	0.1667	0.1612	0.1461	0.1238	0.0982	0.0728	0.0057	0.0	0.0	0.0
300	0.1667	0.1586	0.1367	0.1067	0.0753	0.0481	0.0010	0.0	0.0	0.0
400	0.1667	0.156	0.128	0.0919	0.0578	0.0318	0.0002	0.0	0.0	0.0
500	0.1667	0.1534	0.1198	0.0792	0.0444	0.0211	0.0	0.0	0.0	0.0
600	0.1667	0.1509	0.1121	0.0683	0.0341	0.0140	0.0	0.0	0.0	0.0
700	0.1667	0.1484	0.1049	0.0589	0.0262	0.0093	0.0	0.0	0.0	0.0
800	0.1667	0.146	0.0982	0.0508	0.0202	0.0061	0.0	0.0	0.0	0.0
900	0.1667	0.1436	0.0919	0.0438	0.0155	0.0041	0.0	0.0	0.0	0.0
1000	0.1667	0.1413	0.0861	0.0377	0.0119	0.0027	0.0	0.0	0.0	0.0
2000	0.1667	0.1198	0.0446	0.0086	0.0008	0.0	0.0	0.0	0.0	0.0
3000	0.1667	0.1015	0.0231	0.0020	0.0	0.0	0.0	0.0	0.0	0.0
4000	0.1667	0.0861	0.0120	0.0004	0.0	0.0	0.0	0.0	0.0	0.0
5000	0.1667	0.0730	0.0063	0.0001	0.0	0.0	0.0	0.0	0.0	0.0
6000	0.1667	0.0619	0.0033	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7000	0.1667	0.0525	0.0017	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8000	0.1667	0.0446	0.0009	0.0	0.0	0.0	0.0	0.0	0.0	0.0

9000	0.1667	0.0378	0.0004	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10000	0.1667	0.0321	0.0002	0.0	0.0	0.0	0.0	0.0	0.0	0.0
...										
∞	0.1667	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 6 Probabilities of the event that the decision by voting contradicts that obtained by summing individual utilities with $\sigma/\mu = 0.9129$

Number of voters n	Asymmetry of vote $\lvert\bar{p}-\frac{1}{2}\rvert$									
	0.0	0.01	0.02	0.03	0.04	0.05	0.1	0.2	0.3	0.4
10	0.2355	0.2352	0.2343	0.2329	0.2309	0.2284	0.2079	0.1369	0.0570	0.0102
20	0.2355	0.2349	0.2331	0.2301	0.226	0.2208	0.1812	0.0762	0.0134	0.0005
30	0.2355	0.2346	0.2318	0.2273	0.2212	0.2135	0.158	0.0430	0.0033	0.0
40	0.2355	0.2343	0.2306	0.2246	0.2165	0.2064	0.1379	0.0245	0.0008	0.0
50	0.2355	0.2339	0.2293	0.2219	0.2118	0.1995	0.1205	0.0141	0.0002	0.0
60	0.2355	0.2336	0.2281	0.2192	0.2073	0.1929	0.1053	0.0082	0.0	0.0
70	0.2355	0.2333	0.2269	0.2166	0.2029	0.1866	0.0920	0.0047	0.0	0.0
80	0.2355	0.233	0.2257	0.214	0.1986	0.1804	0.0805	0.0028	0.0	0.0
90	0.2355	0.2327	0.2245	0.2114	0.1944	0.1744	0.0704	0.0016	0.0	0.0
100	0.2355	0.2324	0.2233	0.2089	0.1902	0.1687	0.0617	0.0009	0.0	0.0
200	0.2355	0.2293	0.2116	0.1851	0.1536	0.1209	0.0168	0.0	0.0	0.0
300	0.2355	0.2262	0.2005	0.1642	0.1242	0.0869	0.0047	0.0	0.0	0.0
400	0.2355	0.2232	0.1901	0.1456	0.1006	0.0627	0.0013	0.0	0.0	0.0
500	0.2355	0.2202	0.1802	0.1293	0.0815	0.0454	0.0004	0.0	0.0	0.0
600	0.2355	0.2173	0.1708	0.1148	0.0662	0.0330	0.0001	0.0	0.0	0.0
700	0.2355	0.2144	0.162	0.102	0.0538	0.0240	0.0	0.0	0.0	0.0
800	0.2355	0.2115	0.1536	0.0906	0.0438	0.0175	0.0	0.0	0.0	0.0
900	0.2355	0.2087	0.1456	0.0806	0.0357	0.0128	0.0	0.0	0.0	0.0
1000	0.2355	0.2059	0.1381	0.0717	0.0291	0.0094	0.0	0.0	0.0	0.0
2000	0.2355	0.1801	0.0817	0.0227	0.0040	0.0004	0.0	0.0	0.0	0.0
3000	0.2355	0.1577	0.0487	0.0074	0.0005	0.0	0.0	0.0	0.0	0.0
4000	0.2355	0.1381	0.0293	0.0024	0.0	0.0	0.0	0.0	0.0	0.0
5000	0.2355	0.1211	0.0177	0.0008	0.0	0.0	0.0	0.0	0.0	0.0
6000	0.2355	0.1062	0.0107	0.0002	0.0	0.0	0.0	0.0	0.0	0.0
7000	0.2355	0.0931	0.0065	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8000	0.2355	0.0817	0.0040	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9000	0.2355	0.0718	0.0024	0.0	0.0	0.0	0.0	0.0	0.0	0.0

10000	0.2355	0.0631	0.0015	0.0	0.0	0.0	0.0	0.0	0.0	0.0
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∞	0.2355	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 7 Probabilities of the event that the decision obtained by voting contradicts that obtained by summing individual utilities with $\sigma/\mu = 3$

Number of voters n	Asymmetry of vote $\lvert\bar{p} - \frac{1}{2}\rvert$									
	0.0	0.01	0.02	0.03	0.04	0.05	0.1	0.2	0.3	0.4
10	0.3976	0.3974	0.3969	0.396	0.3948	0.3933	0.3807	0.3341	0.2699	0.2041
20	0.3976	0.3972	0.3961	0.3943	0.3918	0.3886	0.3633	0.2819	0.1937	0.1211
30	0.3976	0.397	0.3953	0.3926	0.3888	0.384	0.3473	0.2417	0.1449	0.0759
40	0.3976	0.3968	0.3946	0.3909	0.3859	0.3796	0.3323	0.2097	0.1108	0.0490
50	0.3976	0.3966	0.3939	0.3893	0.383	0.3752	0.3184	0.1835	0.0858	0.0321
60	0.3976	0.3964	0.3931	0.3876	0.3801	0.3708	0.3055	0.1616	0.0671	0.0213
70	0.3976	0.3962	0.3923	0.3859	0.3773	0.3666	0.2933	0.143	0.0529	0.0143
80	0.3976	0.396	0.3916	0.3843	0.3745	0.3625	0.2819	0.127	0.0419	0.0096
90	0.3976	0.3958	0.3908	0.3827	0.3718	0.3584	0.2712	0.1132	0.0333	0.0065
100	0.3976	0.3956	0.3901	0.3811	0.369	0.3544	0.2611	0.1011	0.0266	0.0044
200	0.3976	0.3938	0.3828	0.3656	0.3435	0.3181	0.1849	0.0356	0.0031	0.0
300	0.3976	0.3919	0.3758	0.3511	0.3208	0.2875	0.1361	0.0135	0.0003	0.0
400	0.3976	0.3901	0.3689	0.3375	0.3003	0.2612	0.1025	0.0053	0.0	0.0
500	0.3976	0.3883	0.3622	0.3248	0.2819	0.2385	0.0782	0.0021	0.0	0.0
600	0.3976	0.3864	0.3558	0.3128	0.2652	0.2185	0.0602	0.0008	0.0	0.0
700	0.3976	0.3846	0.3495	0.3015	0.2499	0.2009	0.0467	0.0003	0.0	0.0
800	0.3976	0.3828	0.3434	0.2909	0.236	0.1852	0.0365	0.0001	0.0	0.0
900	0.3976	0.3809	0.3375	0.2808	0.2231	0.1711	0.0286	0.0	0.0	0.0
1000	0.3976	0.3792	0.3318	0.2713	0.2113	0.1584	0.0225	0.0	0.0	0.0
2000	0.3976	0.3623	0.2819	0.1977	0.1288	0.0785	0.0022	0.0	0.0	0.0
3000	0.3976	0.3465	0.2428	0.1492	0.0828	0.0415	0.0002	0.0	0.0	0.0
4000	0.3976	0.3317	0.2113	0.115	0.0547	0.0226	0.0	0.0	0.0	0.0
5000	0.3976	0.318	0.1853	0.0897	0.0367	0.0126	0.0	0.0	0.0	0.0
6000	0.3976	0.3051	0.1635	0.0707	0.0249	0.0070	0.0	0.0	0.0	0.0
7000	0.3976	0.2932	0.1449	0.0561	0.0170	0.0040	0.0	0.0	0.0	0.0
8000	0.3976	0.282	0.1289	0.0447	0.0117	0.0022	0.0	0.0	0.0	0.0
9000	0.3976	0.2713	0.115	0.0358	0.0081	0.0013	0.0	0.0	0.0	0.0

10000	0.3976	0.2613	0.1029	0.0288	0.0056	0.0007	0.0	0.0	0.0	0.0
⋮										
∞	0.3976	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0